

BALAGANSK, Russian FederationWMO: **306120**Lat: **54.000N**Lon: **103.067E**Elev: **427**StdP: **96.30**Time Zone: **8.00 (E08)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-38.4	-35.6	-41.2	0.1	-38.1	-38.4	0.1	-35.5	7.5	-14.7	6.0	-12.9	1.3	280	0.633

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.2	28.5	18.5	26.7	18.0	25.0	17.5	20.4	25.5	19.4	24.2	18.5	23.2	2.5	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.7	14.2	22.8	17.7	13.4	21.7	16.7	12.5	20.8	60.9	25.4	57.4	24.3	54.3	23.2	24.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.3	6.2	5.3	DB	-39.1	31.7	4.1	1.5	-42.1	32.8	-44.5	33.6	-46.9	34.5	-49.9	35.6	(4)
(5)			WB	-39.0	22.0	3.9	1.1	-41.8	22.8	-44.1	23.4	-46.3	24.1	-49.1	24.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	-0.9	-23.1	-20.1	-10.5	1.2	8.4	16.1	18.9	16.3	8.9	1.0	-10.0	-19.8	(6)
(7)		DBStd	15.58	7.06	7.08	6.76	4.50	4.27	3.83	2.79	3.30	3.99	4.65	7.15	7.36	(7)
(8)		HDD10.0	4723	1026	842	635	264	82	3	0	2	67	279	601	923	(8)
(9)		HDD18.3	7119	1284	1076	893	513	307	88	27	80	282	537	851	1181	(9)
(10)		CDD10.0	727	0	0	0	1	33	184	277	196	35	1	0	0	(10)
(11)		CDD18.3	81	0	0	0	0	1	19	44	16	1	0	0	0	(11)
(12)		CDH23.3	761	0	0	0	0	18	236	364	137	6	0	0	0	(12)
(13)		CDH26.7	154	0	0	0	0	2	52	78	21	1	0	0	0	(13)
(14)	Wind	WSAvg	2.3	1.8	1.8	2.2	2.8	2.9	2.6	2.3	2.3	2.5	2.5	2.3	1.9	(14)
(15)	Precipitation	PrecAvg	337	10	7	6	14	26	51	66	68	46	17	13	14	(15)
(16)		PrecMax	515	17	15	15	30	58	98	118	181	120	36	26	37	(16)
(17)		PrecMin	190	2	1	1	4	5	9	18	19	11	2	4	2	(17)
(18)		PrecStd	79	4	4	4	7	13	26	24	37	22	8	6	8	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-4.3	-1.7	4.9	17.0	26.0	30.8	31.0	28.9	23.5	14.4	4.8	-2.6	(19)
(20)			MCWB	-5.2	-3.0	1.1	8.6	14.4	18.0	19.1	18.6	15.9	9.8	2.5	-3.8	(20)
(21)		2%	DB	-7.5	-5.6	2.7	13.6	22.4	28.1	28.8	26.6	20.0	11.6	2.6	-4.9	(21)
(22)			MCWB	-8.2	-6.7	-0.1	6.5	12.5	17.5	19.4	18.4	14.3	7.8	0.7	-5.8	(22)
(23)		5%	DB	-10.5	-8.0	1.0	10.9	19.5	26.1	27.1	24.6	17.8	9.4	1.3	-7.1	(23)
(24)			MCWB	-11.0	-8.8	-1.2	5.0	10.9	16.8	18.9	17.6	13.0	6.3	-0.3	-7.7	(24)
(25)		10%	DB	-13.0	-10.3	-0.6	8.3	16.8	24.0	25.3	22.9	15.7	7.5	-0.2	-9.6	(25)
(26)			MCWB	-13.4	-11.0	-2.4	3.7	9.5	16.0	18.3	17.1	11.7	4.9	-1.6	-10.1	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-5.0	-2.8	1.8	9.0	15.1	20.4	22.0	21.1	16.7	10.3	3.0	-3.5	(27)
(28)			MCDWB	-4.5	-1.5	3.9	15.4	24.5	26.3	27.1	25.4	21.7	13.9	4.6	-2.6	(28)
(29)		2%	WB	-8.2	-6.7	0.2	7.2	13.2	19.0	20.9	19.3	15.0	8.2	1.0	-5.8	(29)
(30)			MCDWB	-7.6	-5.6	2.2	12.7	20.8	24.9	26.2	24.1	19.6	10.9	2.2	-5.0	(30)
(31)		5%	WB	-11.0	-8.8	-1.0	5.4	11.6	18.0	20.0	18.4	13.5	6.5	-0.2	-7.7	(31)
(32)			MCDWB	-10.5	-8.1	1.0	10.3	18.3	24.1	24.9	23.4	17.1	9.0	1.2	-7.1	(32)
(33)		10%	WB	-13.4	-11.0	-2.5	3.9	10.1	16.9	19.2	17.5	12.1	5.1	-1.5	-10.0	(33)
(34)			MCDWB	-12.9	-10.3	-0.6	8.3	15.9	22.3	23.5	22.2	15.2	7.2	-0.2	-9.5	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.5	9.0	10.6	10.1	12.0	11.5	10.2	9.9	9.4	8.0	7.2	7.4	(35)
(36)			MCDBR	8.5	10.5	10.1	14.4	16.7	15.4	13.7	13.4	12.7	10.6	6.6	8.6	(36)
(37)			MCWBR	7.9	9.6	7.9	8.4	8.5	6.7	5.8	6.4	7.3	7.4	5.4	8.0	(37)
(38)		5% WB	MCDWR	8.3	10.3	9.5	13.1	15.1	12.5	11.0	11.7	11.8	10.1	6.3	8.5	(38)
(39)			MCWBR	7.8	9.5	7.7	8.1	8.2	6.0	5.5	6.4	7.3	7.3	5.3	7.9	(39)
(40)	Clear-Sky Solar Irradiance	taub		0.282	0.308	0.337	0.379	0.426	0.429	0.428	0.391	0.345	0.326	0.301	0.280	(40)
(41)		taud		2.030	2.020	2.044	2.195	2.121	2.221	2.259	2.346	2.431	2.351	2.139	2.007	(41)
(42)		Ebn at Noon		637	755	824	844	819	822	815	827	826	750	614	543	(42)
(43)		Edh at Noon		73	104	128	127	145	133	126	109	88	75	66	61	(43)
(44)	All-Sky Solar Radiation	RadAvg		1.09	2.18	3.80	4.92	5.46	6.01	5.44	4.62	3.22	1.94	1.03	0.74	(44)
(45)		RadStd		0.06	0.10	0.15	0.34	0.45	0.37	0.49	0.27	0.28	0.15	0.09	0.07	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	-2.48	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page